

ER Diagram — Emotion Detection and Learning Support Engine

Date	15 July 2026
Team ID	XXXX
Project Name	Emotion Detection Learning Support Engine

Overview

This entity-relationship diagram describes the planned relational database schema for persisting user accounts and emotion-detection session records. The current application implementation uses CSV-based logging (emotion_response_examples.csv, emotion_response_mapping.csv) for simplicity and rapid development; this schema represents the structure a full MySQL backend would use for multi-user, persistent deployment.

Entities

USERS

Field	Type	Key
email	VARCHAR	Primary Key (PK)
name	VARCHAR	
password	VARCHAR (hashed)	
role	VARCHAR	
login_count	INT	
created_at	DATETIME	

EMOTION_RECORDS

Field	Type	Key
record_id	INT	Primary Key (PK)
email	VARCHAR	Foreign Key (FK) → Users.email
field	VARCHAR	
input_text	TEXT	
predicted_emotion	VARCHAR	
secondary_emotion	VARCHAR	
confidence_score	FLOAT	
model_used	VARCHAR	
ai_response	TEXT	
response_type	VARCHAR	
emotion_scores	JSON / TEXT	
timestamp	DATETIME	
csv_logged	BOOLEAN	

Relationship

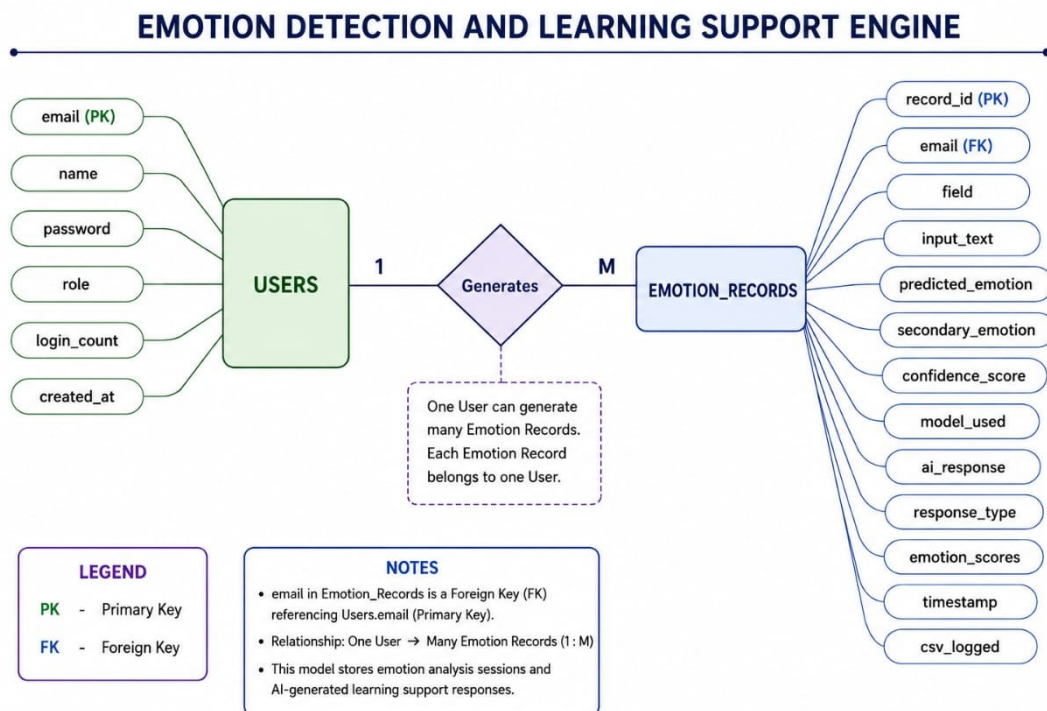
USERS (1) → (M) EMOTION_RECORDS

- One User can generate many Emotion Records.
- Each Emotion Record belongs to exactly one User.
- email in EMOTION_RECORDS is a Foreign Key referencing Users.email (Primary Key).

Notes

- This model stores emotion analysis sessions and AI-generated learning support responses.
- predicted_emotion and secondary_emotion correspond to the primary and mixed-emotion outputs shown in the application UI.
- model_used distinguishes which model (BiLSTM or BERT) produced a given record, supporting the app's side-by-side model comparison feature.
- csv_logged is a boolean flag indicating whether a record was also written to the CSV logs, supporting the transition period between file-based and database-backed storage.

Diagram



Legend

- PK — Primary Key
- FK — Foreign Key